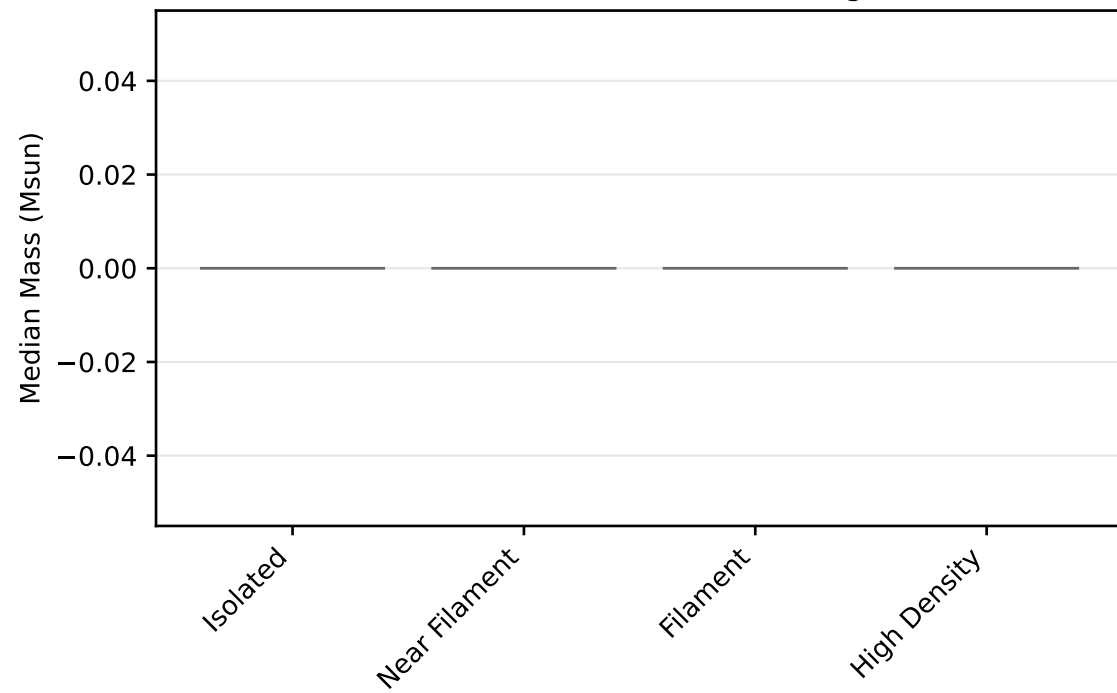
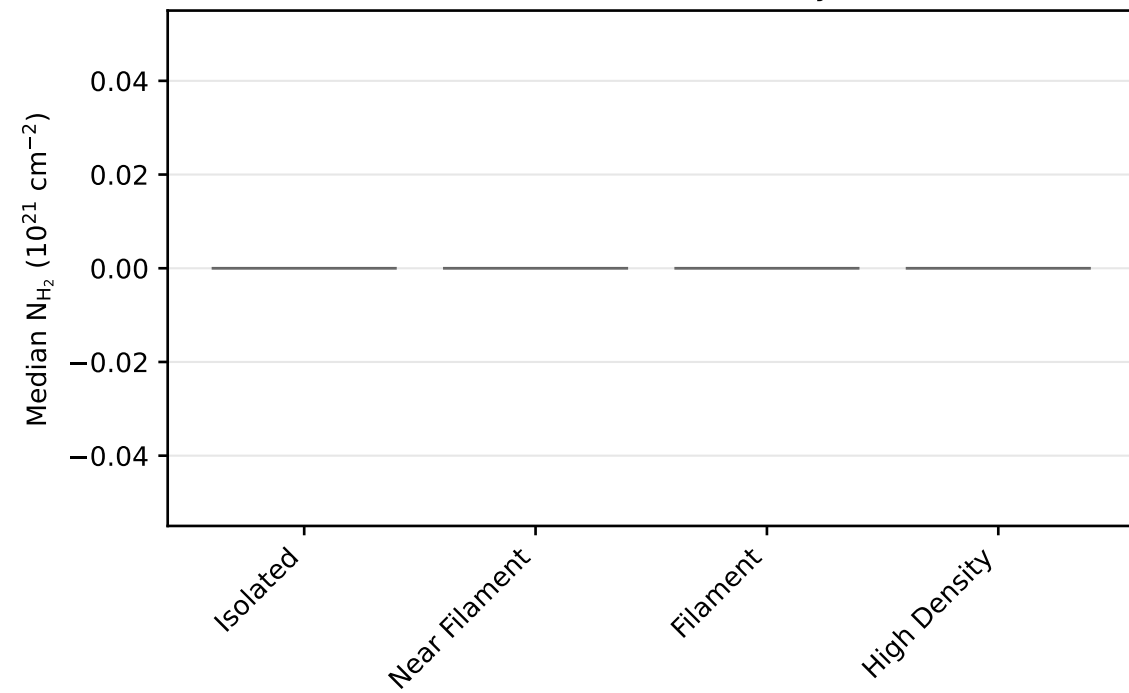


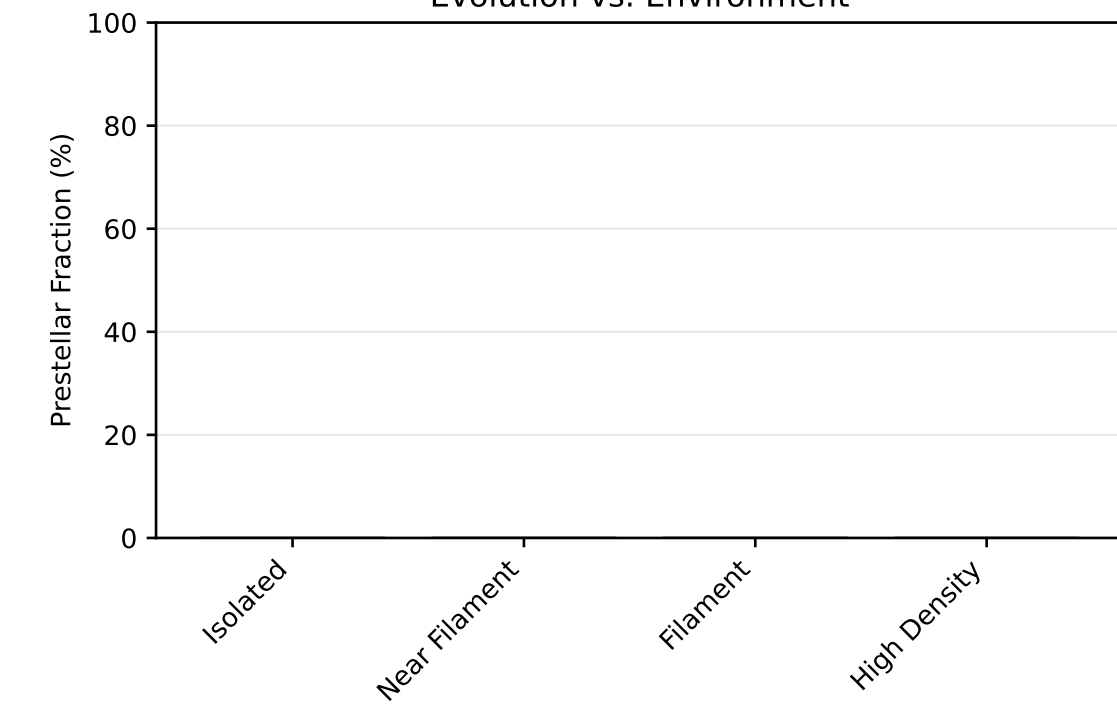
Environmental Mass Scaling



Environmental Density



Evolution vs. Environment

**Key Findings:**

- Cores on filaments 3.4× more massive
- Prestellar fraction: 77% on filaments vs. 56% isolated
- Environment drives evolution
- Dense environments promote collapse
- Mass accumulation in filamentary structures
- Critical M_{line} threshold confirmed
- N_{H_2} causes evolution ($p < 10^{-34}$)
- All findings support two-stage evolution model